



IILM UNIVERSITY

Greater Noida, Uttar Pradesh

An Internship Presentation On

FACE RECOGNITION USING PYTHON AND OPENCV

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Program & Semester	B.Tech CSE (Batch 2024-28)
Organization	Codec Technologies Pvt. Ltd.

Internship Overview

Organization	Codec Technologies Pvt. Ltd. — a Google for Education Partner, offering AICTE & ICAC approved internship programs
Role	Python Developer Intern
Duration	1 month (09/08/2026 to 09/09/2026)
Supervisor	Dr. Anurag Shrivastava, Program Manager
Domain	Computer Vision & Image Processing

Project Overview

A computer-vision-based application developed to detect and recognize human faces from images, combining image-processing techniques with machine-learning-based recognition to build an automated identification system.

Two Core Techniques

Haar Cascade Classifier

Used for face detection — scans an image and locates the position of faces using scale factor, minimum neighbors, and minimum size parameters.

LBPH Face Recognizer

Local Binary Pattern Histogram — trained on grayscale facial images to recognize identities and predict the label of a detected face.

Problem Statement

Existing Problem

Identifying individuals from photographs manually requires visually inspecting images and comparing facial characteristics — this becomes inefficient as the number of images grows.

The System Needs To

- Read an input image
- Detect faces present in the image
- Extract the detected facial regions
- Compare facial features against a prepared training dataset
- Predict the identity label of the detected face
- Display the recognition result

Project Objectives

Primary Objective

Develop a Python-based face-recognition prototype capable of detecting faces in images and predicting their identity using a trained LBPH model.

Specific Objectives

- Understand the fundamentals of computer vision
- Implement face detection using OpenCV
- Organize facial images into a structured dataset
- Extract facial regions and generate labels
- Store training information in CSV format
- Preprocess images into grayscale format
- Train an LBPH face-recognition model
- Detect, predict, and display the recognition result

Technologies & Tools Used

Python

Primary programming language

OpenCV

Image reading, processing, detection & recognition

Haar Cascade Classifier

Detects the location of faces in images

LBPH Recognizer

Trains and predicts facial identity

NumPy

Numerical operations & label arrays

Pandas

Reads the CSV training information

CSV

Stores training image paths and labels

Argparse

Supplies the test image via command line

GitHub

Hosts and shares the project source code

System Architecture

The system separates “where is the face?” from “which trained identity does this face most closely match?” through two sequential stages.

Stage 1 Face Detection



Stage 2 Face Recognition



Training images are organized by person, faces are detected and cropped, and grayscale crops with labels train the LBPH model. A test image then goes through the same detection step before the trained model predicts its label.

Methodology

1

Requirement Analysis

2

Dataset Organization

3

Face Detection

4

Face Extraction

5

Label Generation

6

CSV Creation

7

Image Preprocessing

8

LBPH Model Training

9

Prediction & Result Display

Scope & Limitations

Current Scope

- Training images organized by person
- Haar Cascade face detection
- Cropping of detected faces
- CSV-based label management
- Grayscale preprocessing & LBPH training
- Bounding-box and label visualization

Limitations

- Works with still images, not continuous video
- Depends heavily on training-dataset quality
- Numerical label needs a separate name mapping
- Haar Cascade is less robust than deep-learning detectors
- No graphical user interface currently
- No database for storing user profiles

Future Scope

- Webcam-based real-time recognition
- Real-time video processing
- Name-based identity mapping
- Graphical user interface
- Database integration & attendance management
- Advanced deep-learning face recognition
- Better handling of lighting and pose variation
- Deployment as a web or desktop application

Expected Outcomes

Technical Outcomes

- Read the training dataset & detect faces
- Generate cropped face images and labels
- Train an LBPH recognition model
- Detect faces in the test image
- Predict the identity label with bounding box

Learning Outcomes

- Python programming & computer vision
- Haar Cascade classifiers & LBPH recognition
- Image preprocessing with OpenCV
- CSV-based data handling with Pandas
- Exception handling & GitHub project management

Conclusion

This project demonstrates how computer vision can reduce manual effort in image-based identification tasks, using a Haar Cascade + LBPH pipeline to detect and recognize faces from a trained dataset.

It provides a foundation for more advanced applications such as attendance systems, identity verification, and real-time face recognition — while, as a basic educational implementation, it is not intended as a high-security biometric solution without further validation and stronger recognition methods.

Thank You

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